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Università degli Studi di Napoli "Federico II"
Master in Economics and Finance (MEF)

ECONOMETRICS

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HOMEWORKS

General guidelines:

- These homeworks are not marked. It is your responsibility to do them. For a selection of them the solution will be explained.
- It is likely that the solutions for most problems can be found somewhere on books and other online resources. Try hard before to give up and consult colleagues, other books, notes, etc.
- There is not a precise order here, although these exercises approximately follow the course structure. It is important that you solve practical (computing) problems, and the more theoretical oriented exercises in parallel. Doing matrix algebra on a computer will make the formalism much clearer.
- The notation follows the slide sets given for classes if not otherwise stated.
- R stands for "R computing software"
- "*Errare humanum est, perseverare autem diabolicum*": please notify typos and errors to e-mail: pcoretto@unisa.it.

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1 Preparatory problems: linear algebra and statistics

Problem 1.1 (Vectors, norms, distances and inner product). Consider the following three vectors in $\mathbb{R}^3$

$$\boldsymbol{x}_1 = \begin{pmatrix} -9 \\ -6 \\ -6 \end{pmatrix} \qquad \boldsymbol{x}_2 = \begin{pmatrix} -4 \\ -2 \\ -5 \end{pmatrix} \qquad \boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 4 \\ 10 \end{pmatrix}$$

Do the following calculations by hand and then check the result using R

1. Compute $d_1(\boldsymbol{x}_1, \boldsymbol{x}_2)$, $d_1(\boldsymbol{x}_2, \boldsymbol{x}_1)$, $d_\infty(\boldsymbol{x}_1, \boldsymbol{x}_3)$
2. Compute $d_1(\boldsymbol{x}_1, \boldsymbol{x}_2)$, $d_1(\boldsymbol{x}_2, \boldsymbol{x}_1)$, $d_\infty(\boldsymbol{x}_1, \boldsymbol{x}_3)$
3. Compute $\|\boldsymbol{x}_1\|_2$ and $\|\boldsymbol{x}_2\|_2$
4. Compute $\langle \boldsymbol{x}_1, \boldsymbol{x}_2 \rangle$
5. Compute the angle (degrees) between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$
6. Show that these 3 vectors are linearly independent
7. Can these 3 vectors make an orthogonal basis for $\mathbb{R}^3$

* * *

Problem 1.2 (Matrix algebra). Consider the following matrices

$$\boldsymbol{A} = \begin{pmatrix} -5 & 8 & 9 \\ -3 & -6 & 3 \\ 1 & 8 & 3 \end{pmatrix} \qquad \boldsymbol{B} = \begin{pmatrix} -9 & -2 & 10 \\ -6 & 5 & -2 \\ -6 & -0 & 6 \\ 4 & 4 & 9 \end{pmatrix}$$

Do the following calculations by hand and then check the result using R

1. Compute $\boldsymbol{B}'$
2. Compute $\boldsymbol{A}\boldsymbol{B}'$
3. Use the previous result to compute $\boldsymbol{B}\boldsymbol{A}\boldsymbol{B}'$
4. Compute $\boldsymbol{B}\boldsymbol{B}'$ and $\boldsymbol{B}'\boldsymbol{B}$

* * *

Problem 1.3 (Linear systems). Consider the following linear system in matrix form

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$$

where

$$\boldsymbol{A} = \begin{pmatrix} -5 & 8 \\ -3 & -6 \\ 1 & 8 \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} 18 \\ 0 \\ 6 \end{pmatrix},$$

1. Write the system in equation form
2. Write the system in vector equation form
3. Does a solution exists? (hint: given a matrix $\boldsymbol{X}$, the rank($\boldsymbol{X}$) can be computed in R with `qr(X)$rank`)
4. Does the system have a unique solution?

* * *

Problem 1.4 (Linear independence). Consider the following linear system in matrix form

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$$

where $\boldsymbol{y} \in \mathbb{R}^n$ and $\boldsymbol{x} \in \mathbb{R}^K$.

1. What is the dimension of $\boldsymbol{A}$?
2. Assume that $\boldsymbol{y}$ is linearly independent to all column vectors of $\boldsymbol{A}$. Show that the system cannot have a unique solution.

* * *

Problem 1.5 (Multivariate sample statistics). Assume $X = (X_1, X_2, X_3)$ is a multivariate random variable in dimension $K = 3$. Suppose that we observe $n = 4$ sample units. Let x_{ij} be the sample value of the variable X_j observed on the ith sample unit. The data matrix $\boldsymbol{X} = (x_{ij})$ is

$$\boldsymbol{X} = \begin{pmatrix} -9 & -2 & 0 \\ -6 & 5 & 1 \\ -6 & -0 & 1 \\ 4 & 4 & 0 \end{pmatrix}.$$

The column vector $\boldsymbol{x}_i \in \mathbb{R}^3$ contains the measurements on the ith unit, and this coincides with the ith row vector $\boldsymbol{x}_i'$, contains measurements of the 3 variables on the ith unit. Do the following calculations by hand and then check the result using R.

1. Compute the sample mean vector $\bar{\boldsymbol{x}}_n$

$$\bar{\boldsymbol{x}}_n = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i = \begin{pmatrix} n^{-1}\sum_{i=1}^{n} x_{i1} \\ n^{-1}\sum_{i=1}^{n} x_{i2} \\ \vdots \\ n^{-1}\sum_{i=1}^{n} x_{ik} \end{pmatrix}.$$

2. Compute the sample covariance matrix

$$\boldsymbol{S}_n \quad = \quad \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)' \quad = \quad \left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i\boldsymbol{x}_i'\right) - \overline{\boldsymbol{x}}_n\,\overline{\boldsymbol{x}}_n{}'$$

3. Let $\boldsymbol{Q} = (q_{rk})$ be a diagonal matrix where $q_{kk} = 1/\sqrt{\boldsymbol{S}_{kk}}$ and $q_{rk} = 0$ for $r \neq k$. Verify that the sample correlation matrix is given by

$$\boldsymbol{R}_n = \boldsymbol{Q}\boldsymbol{S}_n\boldsymbol{Q},$$

and compute it.

* * *

Problem 1.6 (Algebra of sample statistics). Consider the data matrix $\boldsymbol{X} = (x_{ij})$ of dimension $n \times K$. Let x_{ij} be the sample value of the variable X_j observed on the ith sample unit. The the ith row, that is the row vector $\boldsymbol{x}_i \in \mathbb{R}^K$, contains measurements of the K variables on the ith unit. Also define the 1-vector

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^K.$$

1. Show that the sample mean vector can be calculated as

$$\overline{x}_n = \frac{1}{n}\boldsymbol{X}'\mathbf{1}$$

2. Let $\boldsymbol{D}$ be the centered data matrix $\boldsymbol{D} = (d_{ij})$ where

$$d_{ij} = x_{ij} - \bar{x}_j$$

That is, $\boldsymbol{D}$ is obtained by subtracting to each element on a column of $\boldsymbol{X}$ its corresponding (column) sample average. This matrix centers all variables on their sample average. Let $\boldsymbol{I}_n$ be the identity matrix of dimension n. Show that $\boldsymbol{D} = \boldsymbol{H}\boldsymbol{X}$ where the *centering matrix*

$$\boldsymbol{H} = \boldsymbol{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'.$$

3. Show that the sample covariance matrix

$$\boldsymbol{S}_n = \frac{1}{n}\boldsymbol{D}'\boldsymbol{D}$$

4. Use R and the formula derived in this exercise to redo the calculations of exercise 1.5.

5. Use the formula derived in this exercise to show that for all vectors $\boldsymbol{u} \in \mathbb{R}^K$ $\boldsymbol{u}'\boldsymbol{S}_n\boldsymbol{u} \geq 0$, which implies that $\boldsymbol{S}_n$ is positive semi-definite.

* * *

Problem 1.7 (Basic exploratory data analysis in R). Let $p_{t,s}$ be the quoted price for the sth stock at time t. Here $t = 1, 2, \ldots, T$, and $s = 1, 2, \ldots, S$. The continuously compounding return over k periods is approximated by

$$r_{t,s} = \log(p_{t,s}) - \log(p_{t-k,s}). \tag{1.7.1}$$

Consider the following data set:

- https://pietro-coretto.github.io/datasets/sp500_0207/readme.txt
- https://pietro-coretto.github.io/datasets/sp500_0207/sp500_0207_prices.csv

1. Import the data set in R.
2. Make sure that the calendar dates are formatted appropriately.
3. Count the number of time periods and the number of stocks sampled in the data set.
4. For each column count the number of missing values.
5. Now consider the `SP500` prices and compute the log-returns based on the formula (1.7.1).
6. Do the same for the stocks named `CLX`, `STJ`, and `STT`.
7. Now consider the returns of `SP500`, `CLX`, `STJ`, and `STT`. Use `cbind()` to form a new data-matrix `Y` whose columns are the returns of these stocks.
8. Compute the sample average return of the stocks in `Y`.
9. Show the histogram of the returns of the stocks in `Y`.
10. Display the parallel box-plots of the returns of the stocks in `Y`.
11. Compute the sample correlation matrix of the returns in `Y`. Comment the result.
12. Show the time series plot for the `SP500` returns within between 2/Jan/2002 and 31/Dec/2003.

* * *

Problem 1.8 (Multivariate Gaussian distribution). Consider a 10-dimensional standard Gaussian vector $Z \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_{10})$.

1. Compute $\Pr\{Z'Z > 5\}$
2. Find the value c such that $\Pr\{Z'Z \leq c\} = 90\%$
3. Compute the probability that a point drawn from $\mathcal{N}(\mathbf{0}, \boldsymbol{I}_{10})$ has an Euclidean distance to the center of the distribution not larger than 12.

* * *

Problem 1.9 (Linear maps of Gaussian variables). Consider a vector of random variables $(X_1, X_2, \ldots, X_p)' \in \mathbb{R}^p$, and $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Consider a constant vector $\boldsymbol{c} = (c_1, c_2, \ldots, c_p)'$ where $\boldsymbol{c} \in \mathbb{R}^p$. Consider a linear combination

$$\boldsymbol{c}'X = \sum_{i=1}^{p} c_i X_i$$

Show that the expectation and variance formula for linear maps implies $\boldsymbol{c}'X \sim \mathcal{N}(\boldsymbol{c}'\mu; \boldsymbol{c}'\boldsymbol{\Sigma}\boldsymbol{c})$

* * *

Problem 1.10 (Linear maps and moments). Consider a vector of random variables $(X_1, X_2, \ldots, X_p)' \in \mathbb{R}^p$, with finite second moment and $\mathrm{Var}[X] = \boldsymbol{\Sigma}$. Consider a constant vector $\boldsymbol{c} = (c_1, c_2, \ldots, c_p)'$ where $\boldsymbol{c} \in \mathbb{R}^p$. Use the calculations of exercise 1.9 to verify that

$$\begin{aligned}
\mathrm{E}\left[\sum_{i=1}^{p} c_i X_i\right] &= \sum_{i=1}^{p} c_i \,\mathrm{E}[X_i] \\
\mathrm{Var}\left[\sum_{i=1}^{p} c_i X_i\right] &= \sum_{i=1}^{p} c_i^2 \,\mathrm{Var}[X_i] + 2\sum_{i=1}^{p}\sum_{j>i} c_i c_j \,\mathrm{Cov}[X_i, X_j]
\end{aligned}$$

* * *

Problem 1.11 (Fantozzi game). Mr. Fantozzi performs a sequential experiment. In each step, the result of the experiment is a score given by a random variable X_i. Starting at time $i = 1$, he tries the experiment sequentially for n steps $i = 1, 2, \ldots, n$. Assume $\mathrm{E}[X_i] = \mu < +\infty$ and $\mathrm{Var}[X_i] = \sigma^2 < +\infty$ for all i. Mr Fantozzi is notoriously an unfortunate romantic bungler, he always performs tasks based on clumsy attempts. Therefore the results X_i cannot be assumed independent from what he did in some of the previous steps. The greatest aspiration for him is to guess the value of the average score

$$Y_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Mr Filini, a close friend of Mr Fantozzi, tells him that in order to guess Y_n there exist an infallible method: repeat the experiment a huge number of times, at some point Y_n will be easy to guess. Mr Fantozzi did this for days and nights! He recomputed the average using past values, but after few days of trials the average score started to take extremely different values after each trial. Use the answer to question 1.10 to guess what is going wrong. What should Mr Fantozzi do?

* * *

Problem 1.12 (Conditional expectations and linear dependence). Consider a pair of random variables X and Z. Assume that $\mathrm{E}[Z|X] = 0$ and $\mathrm{E}[|X|] < +\infty$. Use the properties of conditional expectations to show that

1. $\mathrm{E}[Z] = 0$
2. $\mathrm{Cov}(X, Z) = 0$

2 Least squares and the CLRM finite sample inference

Problem 2.1 (Data matrix and sample averages). Consider the data matrix $\boldsymbol{X} \in \mathbb{R}^{n\times K}$. A Let $\boldsymbol{X} = (x_{ij})$ be the general form of $\boldsymbol{X}$. Rows and columns of the data matrix are organized as explained in exercise 1.5.

1. Verify that the general form of $\boldsymbol{X}'\boldsymbol{X} = (\sum_{i=1}^{n} x_{ij}x_{ik})$ where $j, k = 1, 2, \ldots, K$
2. Verify that $\boldsymbol{X}'\boldsymbol{X} = \sum_{i=1}^{n} \boldsymbol{x}_i\boldsymbol{x}_i'$
3. Argue why $\boldsymbol{X}'\boldsymbol{X}$ is real-valued and symmetric
4. Based on the previous points consider

$$\frac{1}{n}\boldsymbol{X}'\boldsymbol{X} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i\boldsymbol{x}_i'$$

   Let $n^{-1}\boldsymbol{X}'\boldsymbol{X} = (m_{jk})$. Explain what these m_{jk} quantities are estimating for $j = k$ and for $j \neq k$

5. Now imagine that $\boldsymbol{X}$ is the design matrix of a regression model, and suppose that your model includes the intercept/or constant term. Therefore, the first column of $\boldsymbol{X}$ is the unit column vector $\mathbf{1} = (1, 1, \ldots, 1)'$. What is the element on the first column and first row of $n^{-1}\boldsymbol{X}'\boldsymbol{X}$.
6. Can you see the connection between $n^{-1}\boldsymbol{X}'\boldsymbol{X}$ and the sample cov matrix of the design matrix?
7. Let $\tilde{\boldsymbol{X}}$ the data matrix obtained from centering the columns of $\boldsymbol{X}$ on their sample means. Consider $n^{-1}\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}} = (\tilde{m}_{jk})$. What does $\tilde{m}_{jk}$ estimate for $j = k$ and for $j \neq k$?

* * *

Problem 2.2 (LS-LFP solution). Consider the simple investment equation

$$\log(I_i) = \beta_1 + \beta_2 r_i + \beta_3 \log(Y_i)$$

I_i=investment, r_i = real interest rate, Y_i = real output. Suppose that you observe $n = 5$ sample units. Collected data:

I	r	Y
7	1.3	7
148	1.9	25
10	0.2	3
8	0.8	122
33	1.9	20

1. Construct the data matrix $\boldsymbol{X}$. Construct the same matrix in R software.
2. Compute $\boldsymbol{X}'\boldsymbol{X}$. Check your calculations using R.
3. Compute the sample mean vector of the regressors $\bar{\boldsymbol{x}} = (\bar{x}_1, \bar{x}_2, \bar{x}_3)'$. Check your calculations using R.
4. Compute the sample variance-covariance matrix of the regressors using R

* * *

Problem 2.3 (LS-LFP and the sample covariance matrix). Consider exercises 2.1 and 1.6. Wlog assume that all columns of $\boldsymbol{X}$ have been centered to have zero sample mean. Moreover recall that for a matrix $\boldsymbol{A} \in \mathbb{R}^{n\times K}$ the $\text{rank}(\boldsymbol{A}) \leq \min\{n, K\}$, and for compatible matrices $\boldsymbol{A}$ and $\boldsymbol{B}$, $\text{rank}(\boldsymbol{AB}) \leq \min\{\text{rank}(\boldsymbol{A}), \text{rank}(\boldsymbol{B})\}$.

1. Use these facts to show that $\det(\boldsymbol{S}_n) > 0$ implies $\text{rank}(\boldsymbol{X}) = K$.
2. Find a counterexample to show that $\text{rank}(\boldsymbol{X}) = K$ does not imply that $\det(\boldsymbol{S}_n) > 0$ (hint: look at the case of the numerical exercise 2.2).

* * *

Problem 2.4 (LS-LFP solution properties). Consider the least square linear fitting problem. Let $\bar{y}$ be the sample mean of the independent variable, and $\bar{\boldsymbol{x}}$ the sample mean vector of the regressors. Moreover assume that CLRM has a constant/intercept term.

1. Show that the predicted values $\hat{\boldsymbol{y}}$ are orthogonal to the residuals $\boldsymbol{e}$, that is $\hat{\boldsymbol{y}}'\boldsymbol{e} = \boldsymbol{0}$.
2. Show that the sample mean of the predicted values $\hat{\boldsymbol{y}}$ equals the sample mean of the observed ys, that is
$$\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \frac{1}{n}\sum_{i=1}^{n}y_i$$

3. Show that the fitted value $\hat{y}_i$ is an unbiased estimate (conditional on $\boldsymbol{X}$) of the regression function.

* * *

Problem 2.5 (Marginal effects and elasticity of regression functions). Consider a vector of regressors $X = (X_1, X_2, \ldots, X_K)'$, and a regression function $f(X; \boldsymbol{\beta})$. Assuming that differentiability holds, the (percentage) elasticity of $f(\cdot)$ wrt X_j holding $X_1, \ldots, X_{j-1}, X_{j+1}, X_K$ constant is defined as

$$100 \times \frac{\partial f(X; \boldsymbol{\beta})}{\partial X_J} \frac{X_j}{f(X; \boldsymbol{\beta})}$$

The latter measures the ratio of the relative (percentage) change in $f(\cdot)$ wrt the relative change in the input X_j, while all other regressors are kept fixed.

1. Given the conditional mean regression function

$$f(X; \boldsymbol{\beta}) = \mathrm{E}[Y \mid X] = \sum_{i=1}^{n} \beta_j X_j$$

   Give an expression for the elasticity wrt to X_j

2. Suppose

$$f(X; \boldsymbol{\beta}) = \mathrm{E}[Y \mid X] = 10 + 5X_2 + 0.2X_3$$

   Compute the elasticity wrt X_3 when $X_2 = 1$ and give a qualitative interpretation of your result. Recompute the same elasticity for $X_2 = 2$ and for $X_2 = 4$.

* * *

Problem 2.6 (Nominal, ordinal, and dummy variables). On a cross-section of n individuals you measured several variables including

`wage`	wage in Eur
`age`	age in years
`yexp`	years of previous work experience
`sex`	=1 if female, =0 otherwise
`ability`	=H if high, =M if medium, =L if low
`education`	=0 if only obtained compulsory education, =1 otherwise

You want to model log(`wage`) as a linear function of all these regressors. Recall that for a discrete regressor X_k (e.g. a categorical/nominal variable), partial effects are computed as the variations in $\mathrm{E}[Y \mid \boldsymbol{X}]$ obtained by changing the level of X_k and holding all other regressors fixed.

1. `ability` is ordinal. How to include this in a regression model? One way to deal with this is to create multiple dummy variables, e.g.

`abilityH`	=1 if `ability`=H, =0 otherwise
`abilityM`	=1 if `ability`=M, =0 otherwise
`abilityL`	=1 if `ability`=L, =0 otherwise

Now consider a data matrix with the constant term, `age`, `yexp`, `yexp`, `sex`,`abilityH`, `abilityM`, `abilityL`, and `education`. Show that the resulting data matrix produces perfect multicollinearity. This is the so called "*dummy trap*".

2. Suppose that your design matrix contains: constant term, `age`, `yexp`, `sex`, `abilityH`, `abilityM`, and `education`. Does perfect multicollinearity still occur? What did you learn from this example?

3. Suppose that your design matrix contains: constant term, `age`, `yexp`, `sex`, `abilityH`, `abilityM`, and `education`. Write down the regression model for log(`wage`). Assume now that the rank condition is satisfied. How do you estimate the partial effect on log(`wage`) when the ability of an individual goes from L to H? What's the partial effect on log(`wage`) when the ability of an individual goes from L to M? What's the partial effect on log(`wage`) when the ability of an individual goes from M to L?

4. On the basis of the previous points, what is the interpretation of the coefficients of multiple dummy variables introduced to recode a categorical/nominal variable?

5. How would you design your regression model to test whether the ability affects the expected log(`wage`) proportionally with the `age`?

* * *

Problem 2.7 (OLS algebra). Data on the three-variable model

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon$$

yield the following results (notice that the model includes a constant):

$$\boldsymbol{X}'\boldsymbol{X} = \begin{pmatrix} 33 & 0 & 0 \\ 0 & 40 & 20 \\ 0 & 20 & 60 \end{pmatrix} \qquad \boldsymbol{X}'\boldsymbol{y} = \begin{pmatrix} 66 \\ 12 \\ 46 \end{pmatrix} \qquad (\boldsymbol{X}'\boldsymbol{X})^{-1} = \begin{pmatrix} \frac{1}{33} & 0 & 0 \\ 0 & 0.03 & -0.01 \\ 0 & -0.01 & 0.02 \end{pmatrix}.$$

Moreover $\sum_{i=1}^{n} (y_i - \overline{y})^2 = 47.5$.

1. What is the sample size?

2. Compute the OLS estimates and check your calculations in R.

3. Show that TSS $= (\sum_{i=1}^{n} y_i^2) - n\overline{y}^2$ and compute $\boldsymbol{y}'\boldsymbol{y}$. Check your calculations in R

4. Show that $\boldsymbol{e}'\boldsymbol{e} = \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X}\boldsymbol{b}$. Estimate s^2.

5. Estimate the standard error of b_2 and test the hypothesis $\beta_2 = 0$ at the 5% significance level. Check your calculations in R.

6. Compute a conditional prediction for y^0, given $x_2^0 = -4$ and $x_3^0 = 2$. Obtain also a 95% confidence interval for this prediction. If the actual value of y^0 turned out to be 12, what could you suspect about this observation?

* * *

Problem 2.8 (Testing and diagnostics). Suppose that assumptions A1–A5 of the CLRM hold and that the following equation was estimated by ordinary least squares using a sample of quarterly data from 1980 to 1999 inclusive ($n = 80$):

$$\widehat{y}_i = \underset{(3.4)}{2.20} + \underset{(0.05)}{1.04}x_{i2} + \underset{(2.2)}{3.48}x_{i3} + \underset{(0.15)}{0.34}x_{i4}$$

Standard errors are in brackets below the OLS estimates. The explained (regression) sum of squares was 112.5 and the residual sum of squares was 19.5.

1. Which of the slope coefficients are significantly different from 0 at the 5% significance level?
2. Calculate the value of the R^2 for this regression.
3. Calculate the value of the "adjusted" R^2 ($\overline{R^2}$).

* * *

Problem 2.9 (Apple data). Consider the following data set:

- https://pietro-coretto.github.io/datasets/apple/readme.txt
- https://pietro-coretto.github.io/datasets/apple/apple.dat

1. Compute the sample mean and median of all variables in DataSet. Variables will be columns of DataSet. Plot the histogram of each variable.
2. Consult help('apply') and help('IQR'). Use the apply function to compute the interquartile range for all variables in your data set.
3. Consult help('pairs'). Make the scatter plot matrix for all variables in your data set. What do you see? pairs() is a powerful graphic display, isn't it?
4. Consult help('cor'). Compute the correlation matrix for all variables. Does this confirm what you have seen in the previous plot?

5. Consider the Cobb-Dougles production function

$$Y = AL^{\alpha}K^{\beta},$$

where as usual K, L are the inputs, and A, α, β describe technology. Write a linear regression model version of the model above.

6. Use `qOut` as a proxy for Y. The dataset does not give L and K in terms of quanties. Use `vLab/pLab` for measuring L, and use `vCap/pCap` for measuring K (note that this is not rigorous because intermediate materials should also be considered). Construct the data matrix $\boldsymbol{X}$ and the vector $\boldsymbol{y}$.

7. Use matrix computing capabilities of R to calculate $\boldsymbol{b}, \boldsymbol{e}, \hat{\boldsymbol{y}}, s^2, R^2$

8. Check that the algebraic properties of the OLS are numerically fulfilled. Do you note something strange in your results?

9. Estimate the partial effects of labour and capital on production. Interprete your estimates.

10. For all inputs estimate the elasticity of the expected output holding the remaining input to their sample average level. Interpret the result.

11. Assume that the CLRM holds. Estimate the variance-covariance matrix of $\boldsymbol{b}$, and extract the estimated standard errors of the coefficients.

12. Assume that the CLRM holds, and assume Gaussianity of the error term. Consult help('qt'). For all coefficients perform the test $\mathrm{H}_0 : \beta_k = 0$ vs $\mathrm{H}_1 : \beta_k \neq 0$.

13. Read help('lm') and run the following code:

```
## Compute the variables not included in the dataset
K      <- DataSet$vCap / DataSet$pCap
L      <- DataSet$vLab / DataSet$pLab
## Estimate the linear model
fit    <- lm( log( DataSet$qOut ) ~ 1 + log(L) + log(K) )
## Summarize the result contained in the lm object
summary(fit)
```

The fit object is a list, to list its elements run names(fit). Moreover, summary(fit) will summarize the content of fit. Check that these results agree with what you computed previously.

14. Now run

```
fit2    <- lm( log( DataSet$qOut ) ~ log(L) + log(K) - 1 )
summary(fit2)
```

Can you see the difference with the previous estimation?

* * *

Problem 2.10 (F-test). Consider the CLRM and the related testing procedures. Consider the linear model estimated in exercise 2.9.

1. Perform the so called "global test" of the model. This is a "default F-test" run by all software doing linear regression. It tests the null hypothesis that all slope coefficients are equal to zero (not including the constant) . In the specific case treated here, you are testing $\mathrm{H}_0 : \alpha = \beta = 0$ against H_1 : "at least one of the two parameter is equal to zero". Write the null hypothesis in terms of the usual $\boldsymbol{R\beta} = \boldsymbol{r}$ linear system.

2. Use R to compute the F statistic and perform the test at 5% significant level.

3. Now consider the fit object that you created in exercise 2.9. Look at summary(fit), and compare your F statistic computed above, with the F-statistic given in summary(fit). What do you see? Yes, as default R performs the global test [Note: summary(fit) is a list that contains the following elements: names(summary(fit)). Therefore you can get the value of the default F-test as summary(fit)$fstatistic].

4. Explain why it does not make sense to include the intercept in the default F-test.

5. Now construct a test for the hypothesis that the production function shows constant returns to scales. Write down the test in terms of the usual $\boldsymbol{R\beta} = \boldsymbol{r}$ linear system. Perform the test at 10% significance level.

6. Use your estimates to predict the output in case K=24000 and L=400000.

* * *

Problem 2.11 (Icecream data). Consider the following data set:

- https://pietro-coretto.github.io/datasets/icecream/readme.txt
- https://pietro-coretto.github.io/datasets/icecream/icecream.csv

1. Estimate the linear model

$$\texttt{cons}_t = \beta_0 + \beta_1 \texttt{income}_t + \beta_2 \texttt{price}_t + \beta_3 \texttt{temp}_t + \varepsilon_t \tag{2.11.1}$$

   You may use the hints given in point 14 of exercise 2.9. Check the default test on the individual coefficients.

2. Compute the residuals from the previous estimate, let e_t be the residual at time t.

3. Plot the residuals against the fitted values. Do you suspect the existence of heteroskedasticity?

4. Construct a matrix called E with 5 columns containing e_t, and its lagged versions e_{t-1}, $e_{t-2}, e_{t-3}, e_{t-4}$. Note that by doing this, you will lose the first 4 observations. Compute the correlation matrix of E. Produce the pairwise scatter plot with pairs(E). What do you observe?

5. Consider the original residual e_t computed in point 2. Produce the time series plot of e_t. What do you observe? How to do you reconcile this evidence with that emerged in point 4?

6. Do you trust the inference you would make based on the CLRM assumptions? Explain your answer.

* * *

Problem 2.12 (Bias and variance of the OLS with non-spherical errors). Consider the assumptions (A1),(A2),(A3) of the CLRM, moreover let (A4-NSE) hold, that is $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}] = \boldsymbol{\Omega}$ (non spherical error).

1. Show that the OLS estimator is unbiased.
2. Show that the variance of the OLS estimator is

$$\mathrm{Var}[\boldsymbol{b} \mid \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\Omega}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}.$$

3. Verify that the proof of the Gauss-Markov theorem breaks down.

* * *

Problem 2.13 (Generalized Linear Regression Model). The Generalized Linear Regression Model (GLRM) assumes: (A1),(A2),(A3) of the CLRM plus

(A4-V): there exists a known function V such that $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}] = \boldsymbol{\Omega} = V(\boldsymbol{X})$.

Remark. (A4-V) is consistent with A4-NSE but it assumes a little bit more: (i) Ω is a function of $\boldsymbol{X}$ which formalizes common violations of assumption (A4); (ii) since V is a known function of the observables $\boldsymbol{X}$, this means that $\boldsymbol{\Omega}$ can potentially be estimated from data. For the moment we treat $\boldsymbol{\Omega}$ as a known varcov matrix.

The "*Generalized Least Square*" Estimator (GLS) is defined as

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} := \arg\min_{\boldsymbol{\beta}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'\boldsymbol{\Omega}^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \tag{2.13.1}$$

1. Give an intuition why $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}$ should solve the problems of the OLS with non-spherical errors.
2. Show that the unique GLS estimator is given by

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{y} \tag{2.13.2}$$

3. Show that the GLS estimator is unbiased under the GLRM assumptions, that is (A1),(A2),(A3), and (A4-V).
4. Show that the variance of the GLS estimator under the GLRM assumptions is

$$\mathrm{Var}[\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} \mid \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{X})^{-1}. \tag{2.13.3}$$

5. Now assume that (A1),(A2),(A3), and (A4) hold, that is the data are generated according to the CLRM. However, you decide to compute $\hat{\boldsymbol{\beta}}_{\text{GLS}}$. What is $\text{Var}[\hat{\boldsymbol{\beta}}_{\text{GLS}} \mid \boldsymbol{X}]$? What do you conclude?

3 Stochastic process and large sample approximations

Problem 3.1 (Convergence in probability). Consider a sequence of random variables $\{X_1, X_2, \ldots\}$ such that $X_n \sim \text{Bernoulli}\left(\frac{1}{n}\right)$ for $n \in \mathbb{N}$. Show that $X_n \xrightarrow{\mathsf{p}} 0$.

* * *

Problem 3.2 (Delta method for univariate iid sequences). Consider a sequence $\{X_i\}$ where $X_i \overset{\mathsf{iid}}{\sim} \text{Exponential}(\lambda)$ for all $i = 1, 2, \ldots$.

1. Let $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$. Use the appropriate LLN and CLT to verify that:
   (i) $\overline{X} \xrightarrow{\mathsf{as}} \lambda^{-1}$,
   (ii) $\sqrt{n}(\overline{X}_n - \lambda^{-1}) \xrightarrow{\mathsf{d}} \mathcal{N}(0, \lambda^{-2})$
2. Now consider $W_n = \overline{X}_n^{-1}$. Use the delta-method, the continuous mapping theorem, and the previous calculations to show that
   (i) $W_n \xrightarrow{\mathsf{as}} \lambda$
   (ii) $\sqrt{n}(W_n - \lambda) \xrightarrow{\mathsf{d}} \mathcal{N}(0, \lambda^2)$

* * *

Problem 3.3 (Delta method for multivariate sequences). Consider a random vector $\boldsymbol{z}_i = (X_i, Y_i)'$, and consider an iid sequence $\{\boldsymbol{z}_i\}$. Assume that $\text{E}[\boldsymbol{z}_i] = \boldsymbol{\mu} = (\mu_x, \mu_y)'$ and $\text{Var}[\boldsymbol{z}_i] = \boldsymbol{\Omega}$ for all $i = 1, 2, \ldots$. Consider marginal sample averages

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i, \quad \text{and} \quad \bar{Y}_n := \frac{1}{n} \sum_{i=1}^n Y_i.$$

1. Define the statistic $W_n = \log(\bar{X}_n / \bar{Y}_n)$.
   (i) Use an appropriate LLN and the continuous mapping theorem to show that
   $$W_n \xrightarrow{\mathsf{as}} W = \log\left(\frac{\mu_x}{\mu_y}\right)$$
   (ii) Use the delta method to find the distribution of $\sqrt{n}(W_n - W)$

2. Now consider the statistic $U_n = 2\bar{X}_n + \bar{Y}_n^2$. A also consider the vector of statistics $\boldsymbol{H}_n = (W_n, U_n)'$. If appropriate, find out the approximate limit distribution of $\boldsymbol{H}_n$.

* * *

Problem 3.4 (Autoregressive linear model). Find the autocovariance and autocorrelation matrices for the following scalar processes called AR(1) (aka "first order autoregressive linear process"):

$$\varepsilon_t = \rho\varepsilon_{t-1} + \upsilon_t,$$

where $|\rho| < 1$ and υ_t is an independent white noise [Hint: recursively substitute $\varepsilon_{t-1} = \rho\varepsilon_{t-2} + \upsilon_{t-1}$ in ε_t, and express ε_t as a function of all its previous realizations, then compute relevant moments].

* * *

Problem 3.5 (Monte Carlo simulation of an AR process). Consider the AR(1) process of exercise 3.4 with $\upsilon_t \overset{\text{iid}}{\sim} \mathsf{T}(10)$, here $\mathsf{T}(g)$ is the Student-t distribution with g degrees of freedom.

1. Set $\rho = 0.1$. Use R to simulate the sequence $\{\varepsilon_t\}$ for $t = 1, 2, \ldots, 1000$. Initial conditions are: $\varepsilon_0 = 1$, $\upsilon_0 = 0$. Plot the obtained time series. Assess the amount of serial correlation with a correlogram obtained using the `acf()` function.
2. Simulate the previous sequence several times and see the plots to assess whether different sample paths look similar or not.
3. Set $\rho = 1$. Simulate the process as before and redo the plots. As before produce serval sample paths. What do you observe?

* * *

Problem 3.6 (Covariance stationarity). Consider the stochastic process $\{X_t\}_{t\geq 0}$,

$$X_t = Z_1 \cos(\omega t) + Z_2 \sin(\omega t)$$

for some fixed ω, Z_1 and Z_2 are independent standard normals. Show that $\{X_t\}_{t\geq 0}$ is covariance stationary.

* * *

Problem 3.7 (Ergodicity). Consider the stochastic process $\{Z_t\}_{t\geq 0}$, where

$$Z_t = DX_t,$$

and $D \sim \text{Uniform}(-10^6, 10^6)$, $X_t \overset{\text{iid}}{\sim} \text{Normal}(0,1)$. Is $\{Z_t\}_{t\geq 0}$ ergodic? It suffices the intuition.

* * *

Problem 3.8 (Exploratory data analysis of time series based on windowed statistics). Note: this problem is also treated in lab-classes). Consider a sample path of some stochastic process $\{\varepsilon_t\}$. A very rude, but straightforward, exploratory method to understand the behavior of the time series is to look at "dynamically" estimated moments. That is, you make an estimate every l time points, using the previous w samples. These procedures have several names: moving estimates, running estimates, rolling estimates, windowing, dynamic smoothing, etc. There exist sophisticated windowing methods, however, here we consider a *crude* one that is still useful to obtain a *descriptive* view of the dynamic behavior in mean/variance. Let N be the sample size, consider the sequence $\{m_T\}$ and $\{v_T^2\}$

$$m_T = \frac{1}{w}\sum_{i=T-w+1}^{T} \varepsilon_i \qquad \text{and} \qquad v_T^2 = \frac{1}{w}\sum_{i=T-w+1}^{T} (\varepsilon_i - m_T)^2,$$

for $T = w, (w+l), (w+2l), \ldots, (w+\delta l)$, where $\delta = (N+w)/l$ (if δ is fractional ts is rounded to the previous integer). The sequences of sampling estimates $\{m_T\}$ and $\{v_T^2\}$ depends on the choice of w and l. The parameter l is called "period" (you estimate mean/variance every l samples), while w is called "window" (how wide you take a *time frame* to form a "local" estimate of mean/variance).

1. Consider the results of the simulation in exercise 3.5 (part 1). Compute the moving estimates for the mean and variance of the process setting $l = 25$ and $w = 50$. Plot the time series of $\{m_T\}$ and $\{v_T^2\}$. Now try alternative settings for l and w and plot the results.
2. Consider the results of the simulation in exercise 3.5 (part 2). Set $l = 25$ and $w = 50$, and plot the time series of $\{m_T\}$ and $\{v_T^2\}$. Try also to vary l and w as before. What do you learn from this experiment?

4 Linear model with predetermined regressors and its large sample inference

Problem 4.1 (Properties of the the linear model with predetermined regressors). Consider the linear model with predetermined regressors.

1. Show that $\mathrm{E}[\varepsilon_i \,|\, \boldsymbol{x}_i] = 0$ implies (B3).
2. Show that $\mathrm{E}[\varepsilon_i \,|\, \{\varepsilon_{i-1}, \ldots, \varepsilon_1, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots, \boldsymbol{x}_1\}] = 0$ implies (B5).
3. Show that $\mathrm{E}[\varepsilon_i \,|\, \boldsymbol{x}_i] = 0$ implies $\mathrm{E}[\varepsilon_i \,|\, \{\varepsilon_{i-1}, \ldots, \varepsilon_1, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots, \boldsymbol{x}_1\}] = 0$

4. Assume $\mathrm{E}[\varepsilon_i \mid \{\varepsilon_{i-1}, \ldots, \varepsilon_1, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots, \boldsymbol{x}_1\}] = 0$. What is your interpretation of the regression function?

5. Assume that the model has a constant term. Show that (B5) implies that $\{\varepsilon_i\}$ is a scalar martingale difference sequence.

* * *

Problem 4.2 (Algebra of $\hat{\boldsymbol{S}}$). Write the general form of the matrix $\hat{\boldsymbol{S}} = n^{-1}\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X}$. Verify that

$$\frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i' = \frac{1}{n}\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X}.$$

* * *

Problem 4.3 (nls80 data). Consider the following data set:

- https://pietro-coretto.github.io/datasets/nls80/readme.txt
- https://pietro-coretto.github.io/datasets/nls80/nls80.RData

This is an RData archive containing a data.frame called NLS80. Inspect the structure of the data set with str(NLS80) (Note: this problem is also treated in lab-classes).

1. Draw an histogram of the variable `iq`. Here `iq` is the population-demeaned IQ scaled to have sample mean = 100. The IQ is taken as a proxy to quantify the ability of an individual. What do you conclude from this histogram?

2. You want to estimate a linear model for explaining the expected wage (actually log(wage)). You also want to test the hypothesis that marginal effect of the years of education on the expected log(wage) depends on the level of ability. In order to do this you consider the model:

$$\begin{aligned}\log(\texttt{wage}_i) =& \beta_1 + \beta_2\texttt{exper}_i + \beta_3\texttt{tenure}_i + \beta_4\texttt{married}_i \\ &+\beta_5\texttt{south}_i + \beta_6\texttt{urban}_i + \beta_7\texttt{black}_i + \beta_8\texttt{educ}_i \\ &+\beta_9\texttt{iq}_i + \beta_{10}\texttt{educ}_i\times\texttt{iq}_i + \varepsilon_i. \end{aligned} \tag{4.3.1}$$

   The last regressor is an interaction term. Use the matrix calculus capabilities of R to compute the OLS estimates.

3. Compute $\widehat{\mathrm{AVar}}(\boldsymbol{b})$, and the asymptotic standard errors of $\boldsymbol{b}$.

4. Compute the usual t-statistic for testing $\mathrm{H}_0 : \beta_k = 0$ for all parameters in the model. Do you reject the null hypothesis at significance level $\alpha = 5\%$?

5. Back to the original problem, can you conclude that the marginal effect of the years of education on the expected log(wage) depends linearly on the level of ability?

6. Now perform the same estimation using the lm() function as follows:

```
## Attach the dataset
attach(NLS80)
## gennerate  the interaction term
educ_iq <-  educ * iq
## Perform OLS
olsfit <- lm(lwage ~ 1 + exper + tenure + married + south + urban + black +
     educ + iq + educ_iq)
sumols <- summary(olsfit)
sumols
```

   What do you note? Do these standard errors coincide with what you computed before?

7. Recompute the standard errors using the formula that you learned for OLS in finite samples. Compare against sumols again. What do you conclude?

8. Remove the interaction term:

$$\begin{aligned}\log(\texttt{wage}_i) =& \beta_1 + \beta_2\texttt{exper}_i + \beta_3\texttt{tenure}_i + \beta_4\texttt{married}_i \\ &+\beta_5\texttt{south}_i + \beta_6\texttt{urban}_i + \beta_7\texttt{black}_i + \beta_8\texttt{educ}_i + \beta_9\texttt{iq}_i + \varepsilon_i. \end{aligned} \tag{4.3.2}$$

   Redo points 3 and 4.

9. Based on the last estimation, predict the expected variation of the *yearly* wage of an individual who decides to spend an additional year in the school.

* * *

Problem 4.4 (Homoscedasticity). Assume (B1)–(B5) plus (B7).

1. Show that $\boldsymbol{b} \xrightarrow{\mathsf{p}} \boldsymbol{\beta}$

2. Show that $\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta}) \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \mathrm{AVar}(\boldsymbol{b}))$, with $\mathrm{AVar}(\boldsymbol{b}) = \sigma^2\boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}$

* * *

Problem 4.5 (Testing nonlinear hypothesis). A researcher consider the following linear model

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i.$$

Suppose that in a sample of $n = 245$ units, the OLS estimation gave

$$\boldsymbol{b} = \begin{pmatrix} 1.4 \\ -0.45 \\ 10 \end{pmatrix}, \qquad \widehat{\mathrm{AVar}}(\boldsymbol{b}) = \begin{pmatrix} 7.5 & 4.9 & 6.1 \\ 4.9 & 5.5 & 6.8 \\ 6.1 & 6.8 & 11.6 \end{pmatrix}.$$

The researcher wants to test the joint null hypothesis:

$$\begin{aligned} 2\beta_1 + 3\beta_2 =& 1 \\ \beta_1^2 + \beta_2^2 =& 2 \end{aligned}$$

1. Write the null hypothesis in a convenient vector form $\boldsymbol{a}(\boldsymbol{\beta}) = \mathbf{0}$.
2. Compute the derivatives matrix $\boldsymbol{A}(\boldsymbol{\beta})$ (Jacobian of $\boldsymbol{a}(\boldsymbol{\beta})$). Use R to check that the full-rank condition of the Wald's nonlinear restriction test is fulfilled at $\boldsymbol{b}$.
3. Compute the Wald's statistic W using R.
4. Perform the test with significance level $\alpha = 10\%$.

5 Identification and IVs

Problem 5.1 (Identification of the CLRM). Consider the assumptions of the CLRM. Let the model be

$$y_i = \beta_1 + \beta_2 x_{i,2} + \beta_3 x_{i,3} + \dots \beta_K x_{i,K} + \varepsilon_i.$$

1. Show that the parameter $\boldsymbol{\beta}$ is identified, indicating precisely which of the CLRM assumptions are needed for the identification.
2. Now assume that the last regressor is omitted, let it be $x_{i,K}$. Consider the model without $x_{i,K}$:

   $$y_i = \beta_1 + \beta_2 x_{i,2} + \beta_3 x_{i,3} + \dots \beta_K x_{i,K-1} + u_i.$$

   Discuss the endogeneity mechanism that may be introduced by the omission. Is the model identifiable?
3. Suppose one estimates the model in point 2 based on OLS. The orthogonal projection theorem implies that that the resulting residual is still orthogonal to the observed regressors. How do you reconcile this result with the devastating effects on the OLS estimator?

* * *

Problem 5.2 (Omission of relevant variables). Consider the data set in exercise 4.3. Now suppose that you ignore the effect of ability on wages and you formulate the following model

$$\begin{aligned} \log(\texttt{wage}_i) =& \beta_1 + \beta_2 \texttt{exper}_i + \beta_3 \texttt{tenure}_i + \beta_4 \texttt{married}_i \\ &+ \beta_5 \texttt{south}_i + \beta_6 \texttt{urban}_i + \beta_7 \texttt{black}_i + \beta_8 \texttt{educ}_i + u_i. \end{aligned} \tag{5.2.1}$$

1. Use the R code of exercise 4.3 to estimate the model (4.3.2). Store the residuals in a vector named resid_1.
2. Modify the previous code to estimate model (5.2.1). Store the residuals. in a vector named resid_2.

3. Compute the sample correlation between resid_1 and resid_2. Also assess the scatter plot of resid_2 against resid_2. Do you see anything interesting?

4. Compute the sample correlation between `iq` and resid_2. Also assess the scatter plot of resid_2 vs `iq`. Do you see anything interesting?

6 Linear model with endogenous/exogenous regressors, GMM estimation and inference

Problem 6.1 (IV and GMM estimator matrix formulation). Consider the "(C)" framework for GMM estimation. Consider the sampling matrices $\boldsymbol{X} \in \mathbb{R}^{n\times K}$ and $\boldsymbol{Z} \in \mathbb{R}^{n\times L}$, and $\boldsymbol{y} \in \mathbb{R}^n$.

1. Verify that:

$$\begin{aligned}
\boldsymbol{s_{xy}} &= \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i &= \frac{1}{n}\boldsymbol{X}'\boldsymbol{y} \\
\boldsymbol{S_{xz}} &= \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{z}_i' &= \frac{1}{n}\boldsymbol{X}'\boldsymbol{Z} \\
\hat{\boldsymbol{S}} &= \frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i' &= \frac{1}{n}\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X}
\end{aligned}$$

where $e_i = y_i - \boldsymbol{z}_i'\hat{\boldsymbol{\delta}}$, with $\hat{\boldsymbol{\delta}}$ denoting the GMM estimator of $\boldsymbol{\delta}$, and

$$\boldsymbol{B} = \begin{pmatrix} e_1^2 & 0 & \cdots & 0 \\ 0 & e_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e_n^2 \end{pmatrix}$$

2. Verify that $\hat{\boldsymbol{\delta}}_{IV} = (\boldsymbol{X}'\boldsymbol{Z})^{-1}\boldsymbol{X}'\boldsymbol{y}$.

3. Suppose you computed $\hat{\boldsymbol{\delta}}_{IV} = (\boldsymbol{X}'\boldsymbol{Z})^{-1}\boldsymbol{X}'\boldsymbol{y}$. Suppose that $\boldsymbol{X}_o$ is a matrix obtained by an arbitrary permutation of the columns of $\boldsymbol{X}$. Now let $\boldsymbol{\delta}_o = (\boldsymbol{X}_o'\boldsymbol{Z})^{-1}\boldsymbol{X}_o'\boldsymbol{y}$. Verify that $\boldsymbol{\delta}_o = \hat{\boldsymbol{\delta}}_{IV}$. [Hint: you may work with a simple case where $L = 3$, $K = 3$, and $\boldsymbol{X}_o$ is obtained by switching the first two columns of $\boldsymbol{X}$]. Therefore the IV (and also the GMM) estimator does not depend on the order of the exogenous variables into $\boldsymbol{X}$.

* * *

Problem 6.2 (Properties of the IV estimator). Consider the "(C)" framework for GMM estimation with $K = L$, and let $\hat{\boldsymbol{\delta}}_{IV}$ be the GMM estimator (aka "IV estimator").

1. Write the estimation error decomposition for $\hat{\boldsymbol{\delta}}_{IV}$.

2. State the (C) assumptions to achieve consistency, and prove that $\hat{\boldsymbol{\delta}}_{IV} \xrightarrow{\mathsf{p}} \boldsymbol{\delta}$.

3. Compute the asymptotic limit distribution of $\sqrt{n}(\hat{\boldsymbol{\delta}}_{IV} - \boldsymbol{\delta})$, and show that its asymptotic variance is $\boldsymbol{\Sigma}_{\boldsymbol{xz}}^{-1}\boldsymbol{S}\boldsymbol{\Sigma}_{\boldsymbol{xz}}^{-1}$.

4. Argue why $(\boldsymbol{X}'\boldsymbol{Z})^{-1}\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{Z})^{-1}$ approximates (consistently) the variance of $\hat{\boldsymbol{\delta}}_{IV}$ for sufficiently large n.

* * *

Problem 6.3 (Homoscedasticity and IV estimation). Consider the "(C)" framework for GMM estimation with $K = L$, and let $\hat{\boldsymbol{\delta}}_{IV}$ be the GMM estimator (that is the "IV estimator"). Additionaly, assume that conditional homoscedasticity holds with respect to the exogenous variables, that is $\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i] = \sigma^2$ for all $i = 1, 2, \ldots, n$.

1. Show that $\boldsymbol{S} = \mathrm{E}[\varepsilon_i^2 \boldsymbol{x}_i \boldsymbol{x}_i'] = \sigma^2 \,\mathrm{E}[\boldsymbol{x}_i \boldsymbol{x}_i'] = \sigma^2 \boldsymbol{\Sigma}_{\boldsymbol{xx}}$

2. Is $\hat{\boldsymbol{S}} = s^2 \frac{1}{n} \sum_{i=1}^n x_i x_i'$ consistent for $\boldsymbol{S}$?

3. Argue why for sufficiently large n the variance of the $\hat{\boldsymbol{\delta}}_{IV}$ is consistently approximated by

$$s^2(\boldsymbol{X}'\boldsymbol{Z})^{-1}(\boldsymbol{X}'\boldsymbol{X})(\boldsymbol{X}'\boldsymbol{Z})^{-1}$$

* * *

Problem 6.4 (Card's data). Consider the following data set:

- https://pietro-coretto.github.io/datasets/card/readme.txt
- https://pietro-coretto.github.io/datasets/card/card.RData

(Note: this problem is also treated in lab-classes). This is an RData archive containing a data.frame called card_dataset. Inspect the structure of the data set. This is a sample of $n = 3010$ working men aged between 24 and 34 who were part of the 1976 wave of the " US National Longitudinal Survey of Young Men". This data set was used to estimate earnings equations in the famous paper:

> Card, D. (1993). "Using geographic variation in college proximity to estimate the return to schooling", *National Bureau of Economic Research*, Working Paper w4483.

1. Import the data set into an R session and inspect data with str(card_dataset)

2. The goal of Card (1993) (and similar paper on wage equations) is to estimate the rate of returns of years of schooling. In order to do this Card (1993) starts with regressing lwage on educ, exper, expersq, black, south, smsa, reg661, reg662, reg663 , reg664, reg665, reg666, reg667, reg668, smsa66. Let call this: Model-1. Use lm() to estimate Model-1 with OLS.

3. Card (1993) argued why educ is not exogenous in determining level of wages. In fact, it is correlated with the ability which is not measured and it would go into the error of Model-1. Therefore, we know that the orhogonality between educ and the error in Model-1 can't be supported because educ is correlated with ability. Card (1993) proposed to use nearc4 as an IV for educ. We want to compute the IV estimator using nearc4 to instrument educ. Construct the $\boldsymbol{Z}$ and $\boldsymbol{X}$ matrices.

4. Compute the IV estimator of Model-1 with nearc4 as an instrument of educ.

5. Compute the approximate asymptotic "*robust*" standard error of the IV estimator. Are these standard errors based on homoscedasticity?

6. Compare the estimated IV coefficient for educ with the one obtained in the previous OLS regression. Does it make sense to also compare the corresponding standard errors?